

A Literature-Implied Cross-Polytope Subcase of a Small-Radii Maximal-Function Future-Work Direction

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Status. This is a lightweight literature-implied status note:

`literature_implied_answer (partial future-work subcase).`

It records a later answer to the $q = 1$ cross-polytope instance of a future-work direction in Niksinski–Wrobel, *Dimension-free estimates for discrete maximal functions and lattice points in high-dimensional spheres and balls with small radii*, arXiv:2503.16952.

Source direction. The source paper defines, in its introduction,

$$B^q(d) = \{x \in \mathbb{R}^d : \sum_{j=1}^d |x_j|^q \leq 1\}, \quad 1 \leq q < \infty.$$

In Remark 3 of the introduction, immediately after Theorems 1.1–1.5 and the discussion of the remaining Stein range, the authors write that their methods should also extend to maximal functions associated with other B^q balls, provided that the supremum is restricted to small radii, and they identify this as future work. This packet treats that as a future-work target rather than as a formal numbered open problem.

Supporting literature. Kosz–Niksinski–Wrobel, *Uniform estimates for Delannoy numbers and dimension-free estimates for discrete maximal functions over cross-polytopes*, arXiv:2604.15844, studies the $q = 1$ case. Its introduction identifies the relevant sets as cross-polytopes, hyperoctahedrons, or ℓ^1 balls, and defines the discrete averages M_R^d over

$$B_R^d = \{x \in \mathbb{R}^d : |x_1| + \cdots + |x_d| \leq R\}.$$

Theorem 1.15 states that, for every $\varepsilon \in (0, 1)$, there is $C(\varepsilon)$ such that for all dimensions d and all $p \in [2, \infty)$,

$$\|S_{*,(0,d^{1-\varepsilon}]}^d f\|_{\ell^p(\mathbb{Z}^d)} \leq C(\varepsilon) \|f\|_{\ell^p(\mathbb{Z}^d)}$$

and consequently

$$\|M_{*,(0,d^{1-\varepsilon}]}^d f\|_{\ell^p(\mathbb{Z}^d)} \leq C(\varepsilon) \|f\|_{\ell^p(\mathbb{Z}^d)}.$$

Thus the full maximal function over B^1 balls has a dimension-free small-radii estimate in the same sense as the future-work direction of arXiv:2503.16952, at least for the cross-polytope subcase.

The same supporting paper also proves complementary results: Theorem 1.14 gives dimension-free estimates for large radii $R \geq cd^{3/2}$ on all $\ell^p(\mathbb{Z}^d)$, $1 < p < \infty$, and Theorem 1.17 gives a dimension-free dyadic maximal estimate for all radii and $p \in [2, \infty)$. The paper cites arXiv:2503.16952 as reference [19] and explicitly says that its proof of Theorem 1.15 follows the methods of that paper, with the new cross-polytope lattice-point count as the input.

Scope limitations. This is not a new proof produced by this run. It is a later literature identification. It does not answer Stein’s original Euclidean-ball question, does not remove the ε loss from arXiv:2503.16952 for Euclidean balls or spheres, and does not cover all q . For cross-polytopes, it still leaves the full non-dyadic middle range between the small radii treated by Theorem 1.15 and the large radii treated by Theorem 1.14.

Search evidence. Local cheap indexes contained no exact prior packet for arXiv:2503.16952 or arXiv:2604.15844. A web/arXiv search for arXiv:2604.15844 identified the cross-polytope paper and its abstract-level claims; the downloaded PDF was then checked directly for Theorems 1.14, 1.15, 1.17 and for its citation of arXiv:2503.16952.

References.

- J. Niksinski and B. Wrobel, *Dimension-free estimates for discrete maximal functions and lattice points in high-dimensional spheres and balls with small radii*, arXiv:2503.16952, v2, 2026.
- D. Kosz, J. Niksinski, and B. Wrobel, *Uniform estimates for Delannoy numbers and dimension-free estimates for discrete maximal functions over cross-polytopes*, arXiv:2604.15844, v1, 2026.